

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405028
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Parapurath Monangat; Naushad et al.

Intake hood system for an HVAC unit

Abstract

An intake hood system for a heating, ventilation, and/or air conditioning (HVAC) unit includes an intake hood having an inlet configured to receive an air flow into the intake hood, a first support configured to engage a first end of an air filter, and a second support configured to engage a second end of the air filter, where the first support is configured to align with a boundary of the inlet. The intake hood system also includes a lock bar configured to couple to the intake hood while abutting the first support and the second support to secure the air filter within the intake hood.

Inventors: Parapurath Monangat; Naushad (Pune, IN), Singh; Gurpreet (Alwar, IN), Patil; Gaurav S. (Pune, IN), Garg; Karan (Pune, IN), McElvany; John L. (Norman, OK), Pinkston; John M. (Carrollton, TX), Ferreira; Josue D. (Balch Springs, TX)

Applicant: Johnson Controls Tyco IP Holdings LLP (Milwaukee, WI)

Family ID: 73228382

Assignee: JOHNSON CONTROLS LIGHT COMMERCIAL IP GMBH (Neuhausen am Rheinfall, CH)

Appl. No.: 18/351279

Filed: July 12, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230358438 A1	Nov. 09, 2023

Related U.S. Application Data

division parent-doc US 16436534 20190610 US 11739978 child-doc US 18351279
us-provisional-application US 62847155 20190513

Publication Classification

Int. Cl.: F24F13/28 (20060101); F24F7/003 (20210101); F24F7/04 (20060101); F24F8/10 (20210101)

U.S. Cl.:

CPC F24F13/28 (20130101); F24F7/003 (20210101); F24F7/04 (20130101); F24F8/10 (20210101);

Field of Classification Search

CPC: F24F (1/035); F24F (13/20); F24F (13/28)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2825500	12/1957	McClean	N/A	N/A
4465499	12/1983	Wawro et al.	N/A	N/A
4713099	12/1986	Schroeder	N/A	N/A
4759196	12/1987	Davis	55/471	F24F 1/0035
4762053	12/1987	Wolfert	N/A	N/A
4843839	12/1988	Davis	N/A	N/A
5512074	12/1995	Hanni et al.	N/A	N/A
6099608	12/1999	Harms	55/471	F04D 29/283
6387164	12/2001	Cheng	N/A	N/A
6902603	12/2004	Wiser, III et al.	N/A	N/A
6916352	12/2004	Sutton	55/483	B01D 46/12
8999029	12/2014	Brandt et al.	N/A	N/A
9623356	12/2016	Gorman	N/A	N/A
2009/0183463	12/2008	Osborn et al.	N/A	N/A
2013/0125520	12/2012	Gorman	N/A	N/A
2016/0076788	12/2015	Krupo et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2018128990	12/2017	WO	N/A

Primary Examiner: Schult; Allen R. B.

Attorney, Agent or Firm: Fletcher Yoder, PC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. Non-Provisional patent application Ser. No. 16/436,534, entitled “INTAKE HOOD SYSTEM FOR AN HVAC UNIT,” filed Jun. 10, 2019, which claims priority to and the benefit of U.S. Provisional

Application Ser. No. 62/847,155, entitled "INTAKE HOOD SYSTEM FOR AN HVAC UNIT," filed May 13, 2019, each of which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

(1) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present techniques, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

(2) A heating, ventilation, and/or air conditioning (HVAC) system may be used to thermally regulate an environment, such as a building, home, or other structure. The HVAC system generally includes an intake system that receives air, such as outdoor air, for circulation within and thermal regulation of the environment. As air is directed through the intake system, the air is directed through a filter that removes dirt, debris, and other particles from the air. In this way, the intake system may provide clean air to the environment after the air passes through the filter. During operation, dirt, debris, and other particles may accumulate on and within the filter, such that the filter may be replaced with a new or clean filter. In some instances, the HVAC system may not be configured for efficient access to the filter. For example, servicing and/or replacing the filter may be difficult for service personnel and may involve removing portions of the intake system or removing the intake system entirely in order to access the filter. As such, service times to replace or clean the filter may be lengthy.

SUMMARY

(3) A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(4) In one embodiment, an intake hood system for a heating, ventilation, and/or air conditioning (HVAC) unit includes an intake hood having an inlet configured to receive an air flow into the intake hood, a first support configured to engage a first end of an air filter, and a second support configured to engage a second end of the air filter, where the first support is configured to align with a boundary of the inlet. The intake hood system also includes a lock bar configured to couple to the intake hood while abutting the first support and the second support to secure the air filter within the intake hood.

(5) In another embodiment, an intake hood system for a heating, ventilation, and/or air conditioning (HVAC) unit includes an intake hood having a first side panel and a second side panel forming an inlet of the intake hood. The first side panel includes a first lateral channel configured to capture a first side of an air filter, and the second side panel includes a second lateral channel configured to capture a second side of the air filter. The intake hood system also includes a first support disposed between the first side panel and the second side panel such that a length of the first support is transverse to corresponding lengths of the first and second lateral channels, where the first support is configured to capture a first end of the air filter and align with the inlet. Additionally, the intake hood system includes a second support disposed between the first side panel and the second side panel and configured to capture a second end of the air filter.

(6) In yet another embodiment, an intake hood system for a heating, ventilation, and/or air conditioning (HVAC) unit includes a first side panel and a second side panel forming an intake hood. The first side panel and the second side panel each have a first inner flange and a second inner flange. The intake hood system also includes a first C-channel configured to interference fit between the first inner flange and the second inner flange of each of the first side panel and the

second side panel and configured to capture a first end of an air filter, and an intermediate bracket configured to interference fit between the first inner flange and the second inner flange of each of the first side panel and the second side panel. The intermediate bracket includes a second C-channel and a third C-channel, where the second C-channel is configured to capture a second end of the air filter, and the third C-channel is configured to capture a first end of an additional air filter. Additionally, the intake hood system includes a fourth C-channel configured to capture a second end of the additional air filter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various aspects of the present disclosure may be better understood upon reading the following detailed description and upon reference to the drawings, in which:
- (2) FIG. 1 is a perspective view of an embodiment of a heating, ventilation, and/or air conditioning (HVAC) system for building environmental management that may employ one or more HVAC units, in accordance with an aspect of the present disclosure;
- (3) FIG. 2 is a perspective view of an embodiment of a packaged HVAC unit, in accordance with an aspect of the present disclosure;
- (4) FIG. 3 is a perspective view of an embodiment of a residential, split HVAC system, in accordance with an aspect of the present disclosure;
- (5) FIG. 4 is a schematic of an embodiment of a vapor compression system that may be used in an HVAC system, in accordance with an aspect of the present disclosure;
- (6) FIG. 5 is a perspective view of an embodiment of an intake hood system coupled to a housing of an HVAC unit, in accordance with an aspect of the present disclosure;
- (7) FIG. 6 is a perspective view of an embodiment of an intake hood system, in accordance with an aspect of the present disclosure;
- (8) FIG. 7 is a perspective view of an embodiment of an intake hood system, illustrating installation of an air filter with a support of the intake hood system, in accordance with an aspect of the present disclosure;
- (9) FIG. 8 is a perspective view of an embodiment of an intake hood system, illustrating installation of an intermediate bracket of the intake hood system, in accordance with an aspect of the present disclosure;
- (10) FIG. 9 is a perspective view of an embodiment of an intake hood system, illustrating installation of an additional air filter of the intake hood system, in accordance with an aspect of the present disclosure;
- (11) FIG. 10 is a perspective view of an embodiment of an intake hood system, illustrating installation of an additional support of the intake hood system, in accordance with an aspect of the present disclosure;
- (12) FIG. 11 is a perspective view of an embodiment of an intake hood system, illustrating installation of lock bars of the intake hood system, in accordance with an aspect of the present disclosure;
- (13) FIG. 12 is an expanded perspective view of an embodiment of the intake hood system, illustrating an installed lock bar of the intake hood system, in accordance with an aspect of the present disclosure;
- (14) FIG. 13 is an expanded perspective view of an embodiment of the intake hood system, illustrating an installed lock bar of the intake hood system, in accordance with an aspect of the present disclosure;
- (15) FIG. 14 is a flow diagram of an embodiment of a process for assembling an intake hood system, in accordance with an aspect of the present disclosure; and

(16) FIG. 15 is a flow diagram of an embodiment of a process for disassembling an intake hood system, in accordance with an aspect of the present disclosure.

DETAILED DESCRIPTION

(17) One or more specific embodiments of the present disclosure will be described below. These described embodiments are only examples of the presently disclosed techniques. Additionally, in an effort to provide a concise description of these embodiments, all features of an actual implementation may not be described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but may nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(18) When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(19) Generally, a heating, ventilation, and/or air conditioning (HVAC) system may be used to thermally regulate an environment, such as a building, home, or other structure. The HVAC system generally includes an intake hood system that receives air, such as outdoor air, for circulation within and thermal regulation of the environment. As air is directed through the intake hood system, the air is directed through an air filter that removes dirt, debris, and other particles from the air. As a result, the intake hood system may provide clean air to the environment after the air passes through the air filter. During operation of the HVAC system, dirt, debris, and other particles may accumulate on and within the air filter, such that the air filter may be periodically replaced with a clean air filter.

(20) In some instances, servicing and/or replacing the air filter may be complicated for service personnel and may involve removing portions of the intake hood system or the HVAC system generally, such as certain panels, or may involve removing the intake hood system entirely. For example, in a shutter-type intake hood arrangement, the air filter may be pushed upwardly and pulled out of the intake hood system. In some embodiments, a hood of the intake hood system may include side panels that are removed to access the air filter. In certain embodiments, the entire hood is removed from an HVAC unit to access the air filter. As such, service times to replace and/or clean the air filter may be lengthy. Additionally, the removal and reinstallation of the panels and intake hood may cause the panels and intake hood to incur wear, such as wear to fasteners and connecting components of the panels and intake hood.

(21) Accordingly, the present disclosure provides systems and methods for an intake hood system generally positioned at an inlet of an HVAC system and configured to enable improved air filter installation and removal. The intake hood system includes air filters configured to remove dirt, debris, and other particles from air directed through the intake hood system. As discussed in detail below, the disclosed techniques enable quick and efficient access to the air filters, such as access by service personnel to quickly service and/or replace the air filters. For example, the intake hood system may include supports, brackets, and/or side panels that capture the air filters. The supports, the brackets, and/or the side panels may be easily disassembled/uninstalled to remove the air filters and assembled/installed to replace the air filters. As such, the systems and methods described herein improve serviceability of the HVAC system.

(22) Turning now to the drawings, FIG. 1 illustrates an embodiment of a heating, ventilation,

and/or air conditioning (HVAC) system for environmental management that may employ one or more HVAC units. As used herein, an HVAC system includes any number of components configured to enable regulation of parameters related to climate characteristics, such as temperature, humidity, air flow, pressure, air quality, and so forth. For example, an “HVAC system” as used herein is defined as conventionally understood and as further described herein. Components or parts of an “HVAC system” may include, but are not limited to, all, some of, or individual parts such as a heat exchanger, a heater, an air flow control device, such as a fan, a sensor configured to detect a climate characteristic or operating parameter, a filter, a control device configured to regulate operation of an HVAC system component, a component configured to enable regulation of climate characteristics, or a combination thereof. An “HVAC system” is a system configured to provide such functions as heating, cooling, ventilation, dehumidification, pressurization, refrigeration, filtration, or any combination thereof. The embodiments described herein may be utilized in a variety of applications to control climate characteristics, such as residential, commercial, industrial, transportation, or other applications where climate control is desired.

(23) In the illustrated embodiment, a building **10** is air conditioned by a system that includes an HVAC unit **12**. The building **10** may be a commercial structure or a residential structure. As shown, the HVAC unit **12** is disposed on the roof of the building **10**; however, the HVAC unit **12** may be located in other equipment rooms or areas adjacent the building **10**. The HVAC unit **12** may be a single package unit containing other equipment, such as a blower, integrated air handler, and/or auxiliary heating unit. In other embodiments, the HVAC unit **12** may be part of a split HVAC system, such as the system shown in FIG. 3, which includes an outdoor HVAC unit **58** and an indoor HVAC unit **56**.

(24) In any case, the HVAC unit **12** may be an air cooled device that implements a refrigeration cycle to provide conditioned air to the building **10**. For example, the HVAC unit **12** may include one or more heat exchangers across which an air flow is passed to condition the air flow before the air flow is supplied to the building. In the illustrated embodiment, the HVAC unit **12** is a rooftop unit (RTU) that conditions a supply air stream, such as environmental air and/or a return air flow from the building **10**. After the air is conditioned, the HVAC unit **12** may supply the conditioned air to the building **10** via ductwork **14** extending throughout the building **10** from the HVAC unit **12**. For example, the ductwork **14** may extend to various individual floors or other sections of the building **10**. In some embodiments, the HVAC unit **12** may include a heat pump that provides both heating and cooling to the building **10**, for example, with one refrigeration circuit implemented to operate in multiple different modes. In other embodiments, the HVAC unit **12** may include one or more refrigeration circuits for cooling an air stream and a furnace for heating the air stream.

(25) A control device **16**, one type of which may be a thermostat, may be used to designate the temperature of the conditioned air. The control device **16** also may be used to control the flow of air through the ductwork **14**. For example, the control device **16** may be used to regulate operation of one or more components of the HVAC unit **12** or other equipment, such as dampers and fans, within the building **10** that may control flow of air through and/or from the ductwork **14**. In some embodiments, other devices may be included in the system, such as pressure and/or temperature transducers or switches that sense the temperatures and pressures of the supply air, return air, and/or the like. Moreover, the control device **16** may include computer systems that are integrated with or separate from other building control or monitoring systems, and even systems that are remote from the building **10**. In some embodiments, the HVAC unit **12** may operate in multiple zones of the building and may be coupled to multiple control devices that each control flow of air in a respective zone. For example, a first control device **16** may control the flow of air in a first zone **17** of the building, a second control device **18** may control the flow of air in a second zone **19** of the building, and a third control device **20** may control the flow of air in a third zone **21** of the building.

(26) FIG. 2 is a perspective view of an embodiment of the HVAC unit **12**. In the illustrated

embodiment, the HVAC unit **12** is a single package unit that may include one or more independent refrigeration circuits and components that are tested, charged, wired, piped, and ready for installation. The HVAC unit **12** may provide a variety of heating and/or cooling functions, such as cooling only, heating only, cooling with electric heat, cooling with dehumidification, cooling with gas heat, or cooling with a heat pump. As described above, the HVAC unit **12** may directly cool and/or heat an air stream provided to the building **10** to condition a space in the building **10**.

(27) As shown in the illustrated embodiment of FIG. 2, a cabinet **24** or enclosure encloses the HVAC unit **12** and provides structural support and protection to the internal components from environmental and other contaminants. In some embodiments, the cabinet **24** may be constructed of galvanized steel and insulated with aluminum foil faced insulation. Rails **26** may be joined to the bottom perimeter of the cabinet **24** and provide a foundation for the HVAC unit **12**. In certain embodiments, the rails **26** may provide access for a forklift and/or overhead rigging to facilitate installation and/or removal of the HVAC unit **12**. In some embodiments, the rails **26** may fit into “curbs” on the roof to enable the HVAC unit **12** to provide air to the ductwork **14** from the bottom of the HVAC unit **12** while blocking elements such as rain from leaking into the building **10**.

(28) The HVAC unit **12** includes heat exchangers **28** and **30** in fluid communication with one or more refrigeration circuits. Tubes within the heat exchangers **28** and **30** may circulate refrigerant, such as R-410A, through the heat exchangers **28** and **30**. The tubes may be of various types, such as multichannel tubes, conventional copper or aluminum tubing, and so forth. Together, the heat exchangers **28** and **30** may implement a thermal cycle in which the refrigerant undergoes phase changes and/or temperature changes as it flows through the heat exchangers **28** and **30** to produce heated and/or cooled air. For example, the heat exchanger **28** may function as a condenser where heat is released from the refrigerant to ambient air, and the heat exchanger **30** may function as an evaporator where the refrigerant absorbs heat to cool an air stream. In other embodiments, the HVAC unit **12** may operate in a heat pump mode where the roles of the heat exchangers **28** and **30** may be reversed. That is, the heat exchanger **28** may function as an evaporator and the heat exchanger **30** may function as a condenser. In further embodiments, the HVAC unit **12** may include a furnace for heating the air stream that is supplied to the building **10**. While the illustrated embodiment of FIG. 2 shows the HVAC unit **12** having two of the heat exchangers **28** and **30**, in other embodiments, the HVAC unit **12** may include one heat exchanger or more than two heat exchangers.

(29) The heat exchanger **30** is located within a compartment **31** that separates the heat exchanger **30** from the heat exchanger **28**. Fans **32** draw air from the environment through the heat exchanger **28**. Air may be heated and/or cooled as the air flows through the heat exchanger **28** before being released back to the environment surrounding the HVAC unit **12**. A blower assembly **34**, powered by a motor **36**, draws air through the heat exchanger **30** to heat or cool the air. The heated or cooled air may be directed to the building **10** by the ductwork **14**, which may be connected to the HVAC unit **12**. Before flowing through the heat exchanger **30**, the conditioned air flows through one or more filters **38** that may remove particulates and contaminants from the air. In certain embodiments, the filters **38** may be disposed on the air intake side of the heat exchanger **30** to prevent contaminants from contacting the heat exchanger **30**.

(30) The HVAC unit **12** also may include other equipment for implementing the thermal cycle. Compressors **42** increase the pressure and temperature of the refrigerant before the refrigerant enters the heat exchanger **28**. The compressors **42** may be any suitable type of compressors, such as scroll compressors, rotary compressors, screw compressors, or reciprocating compressors. In some embodiments, the compressors **42** may include a pair of hermetic direct drive compressors arranged in a dual stage configuration **44**. However, in other embodiments, any number of the compressors **42** may be provided to achieve various stages of heating and/or cooling. As may be appreciated, additional equipment and devices may be included in the HVAC unit **12**, such as a solid-core filter drier, a drain pan, a disconnect switch, an economizer, pressure switches, phase monitors, and

humidity sensors, among other things.

(31) The HVAC unit **12** may receive power through a terminal block **46**. For example, a high voltage power source may be connected to the terminal block **46** to power the equipment. The operation of the HVAC unit **12** may be governed or regulated by a control board or controller **48**. The control board **48** may include control circuitry connected to a thermostat, sensors, and alarms. One or more of these components may be referred to herein separately or collectively as the control device **16**. The control circuitry may be configured to control operation of the equipment, provide alarms, and monitor safety switches. Wiring **49** may connect the control board **48** and the terminal block **46** to the equipment of the HVAC unit **12**.

(32) FIG. **3** illustrates a residential heating and cooling system **50**, also in accordance with present techniques. The residential heating and cooling system **50** may provide heated and cooled air to a residential structure, as well as provide outside air for ventilation and provide improved indoor air quality (IAQ) through devices such as ultraviolet lights and air filters. In the illustrated embodiment, the residential heating and cooling system **50** is a split HVAC system. In general, a residence **52** conditioned by a split HVAC system may include refrigerant conduits **54** that operatively couple the indoor unit **56** to the outdoor unit **58**. The indoor unit **56** may be positioned in a utility room, an attic, a basement, and so forth. The outdoor unit **58** is typically situated adjacent to a side of residence **52** and is covered by a shroud to protect the system components and to prevent leaves and other debris or contaminants from entering the unit. The refrigerant conduits **54** transfer refrigerant between the indoor unit **56** and the outdoor unit **58**, typically transferring primarily liquid refrigerant in one direction and primarily vaporized refrigerant in an opposite direction.

(33) When the system shown in FIG. **3** is operating as an air conditioner, a heat exchanger **60** in the outdoor unit **58** serves as a condenser for re-condensing vaporized refrigerant flowing from the indoor unit **56** to the outdoor unit **58** via one of the refrigerant conduits **54**. In these applications, a heat exchanger **62** of the indoor unit **56** functions as an evaporator. Specifically, the heat exchanger **62** receives liquid refrigerant, which may be expanded by an expansion device, and evaporates the refrigerant before returning it to the outdoor unit **58**.

(34) The outdoor unit **58** draws environmental air through the heat exchanger **60** using a fan **64** and expels the air above the outdoor unit **58**. When operating as an air conditioner, the air is heated by the heat exchanger **60** within the outdoor unit **58** and exits the unit at a temperature higher than it entered. The indoor unit **56** includes a blower or fan **66** that directs air through or across the indoor heat exchanger **62**, where the air is cooled when the system is operating in air conditioning mode. Thereafter, the air is passed through ductwork **68** that directs the air to the residence **52**. The overall system operates to maintain a desired temperature as set by a system controller. When the temperature sensed inside the residence **52** is higher than the set point on the thermostat, or a set point plus a small amount, the residential heating and cooling system **50** may become operative to refrigerate additional air for circulation through the residence **52**. When the temperature reaches the set point, or a set point minus a small amount, the residential heating and cooling system **50** may stop the refrigeration cycle temporarily.

(35) The residential heating and cooling system **50** may also operate as a heat pump. When operating as a heat pump, the roles of heat exchangers **60** and **62** are reversed. That is, the heat exchanger **60** of the outdoor unit **58** will serve as an evaporator to evaporate refrigerant and thereby cool air entering the outdoor unit **58** as the air passes over outdoor the heat exchanger **60**. The indoor heat exchanger **62** will receive a stream of air blown over it and will heat the air by condensing the refrigerant.

(36) In some embodiments, the indoor unit **56** may include a furnace system **70**. For example, the indoor unit **56** may include the furnace system **70** when the residential heating and cooling system **50** is not configured to operate as a heat pump. The furnace system **70** may include a burner assembly and heat exchanger, among other components, inside the indoor unit **56**. Fuel is provided

to the burner assembly of the furnace system **70** where it is mixed with air and combusted to form combustion products. The combustion products may pass through tubes or piping in a heat exchanger, separate from heat exchanger **62**, such that air directed by the blower **66** passes over the tubes or pipes and extracts heat from the combustion products. The heated air may then be routed from the furnace system **70** to the ductwork **68** for heating the residence **52**.

(37) FIG. **4** is an embodiment of a vapor compression system **72** that may be used in any of the systems described above. The vapor compression system **72** may circulate a refrigerant through a circuit starting with a compressor **74**. The circuit may also include a condenser **76**, an expansion valve(s) or device(s) **78**, and an evaporator **80**. The vapor compression system **72** may further include a control panel **82** that has an analog to digital (A/D) converter **84**, a microprocessor **86**, a non-volatile memory **88**, and/or an interface board **90**. The control panel **82** and its components may function to regulate operation of the vapor compression system **72** based on feedback from an operator, from sensors of the vapor compression system **72** that detect operating conditions, and so forth.

(38) In some embodiments, the vapor compression system **72** may use one or more of a variable speed drive (VSDs) **92**, a motor **94**, the compressor **74**, the condenser **76**, the expansion valve or device **78**, and/or the evaporator **80**. The motor **94** may drive the compressor **74** and may be powered by the variable speed drive (VSD) **92**. The VSD **92** receives alternating current (AC) power having a particular fixed line voltage and fixed line frequency from an AC power source, and provides power having a variable voltage and frequency to the motor **94**. In other embodiments, the motor **94** may be powered directly from an AC or direct current (DC) power source. The motor **94** may include any type of electric motor that may be powered by a VSD or directly from an AC or DC power source, such as a switched reluctance motor, an induction motor, an electronically commutated permanent magnet motor, or another suitable motor.

(39) The compressor **74** compresses a refrigerant vapor and delivers the vapor to the condenser **76** through a discharge passage. In some embodiments, the compressor **74** may be a centrifugal compressor. The refrigerant vapor delivered by the compressor **74** to the condenser **76** may transfer heat to a fluid passing across the condenser **76**, such as ambient or environmental air **96**. The refrigerant vapor may condense to a refrigerant liquid in the condenser **76** as a result of thermal heat transfer with the environmental air **96**. The liquid refrigerant from the condenser **76** may flow through the expansion device **78** to the evaporator **80**.

(40) The liquid refrigerant delivered to the evaporator **80** may absorb heat from another air stream, such as a supply air stream **98** provided to the building **10** or the residence **52**. For example, the supply air stream **98** may include ambient or environmental air, return air from a building, or a combination of the two. The liquid refrigerant in the evaporator **80** may undergo a phase change from the liquid refrigerant to a refrigerant vapor. In this manner, the evaporator **80** may reduce the temperature of the supply air stream **98** via thermal heat transfer with the refrigerant. Thereafter, the vapor refrigerant exits the evaporator **80** and returns to the compressor **74** by a suction line to complete the cycle.

(41) In some embodiments, the vapor compression system **72** may further include a reheat coil in addition to the evaporator **80**. For example, the reheat coil may be positioned downstream of the evaporator relative to the supply air stream **98** and may reheat the supply air stream **98** when the supply air stream **98** is overcooled to remove humidity from the supply air stream **98** before the supply air stream **98** is directed to the building **10** or the residence **52**.

(42) It should be appreciated that any of the features described herein may be incorporated with the HVAC unit **12**, the residential heating and cooling system **50**, or other HVAC systems.

Additionally, while the features disclosed herein are described in the context of embodiments that directly heat and cool a supply air stream provided to a building or other load, embodiments of the present disclosure may be applicable to other HVAC systems as well. For example, the features described herein may be applied to mechanical cooling systems, free cooling systems, chiller

systems, or other heat pump or refrigeration applications.

(43) The description above with reference to FIGS. **1-4** is intended to be illustrative of the context of the present disclosure. The techniques of the present disclosure may be incorporated with any or all of the features described above, as well as other systems not described above. In particular, as will be discussed in more detail below, the present disclosure provides techniques that enable efficient installation/replacement of air filters of an HVAC system. For example, an intake hood system of the HVAC system may be configured for efficient assembly and/or disassembly to enable efficient installation/replacement of the air filters.

(44) To help illustrate, FIG. **5** is a perspective view an HVAC system **100** having an intake hood system **102**. The illustrated HVAC system **100** may include embodiments or components of the HVAC unit **12** shown in FIG. **1**, embodiments or components of the residential heating and cooling system **50** shown in FIG. **3**, a rooftop unit (RTU), or any other suitable HVAC system. For example, the intake hood system **102** may be an intake hood system of an embodiment of the HVAC unit **12**. The intake hood system **102** is configured to receive air into the HVAC system **100**. For example, the HVAC system **100** may be positioned outdoors, and the intake hood system **102** may receive unfiltered, outdoor air for conditioning by the HVAC system **100**. In certain embodiments, the HVAC system **100** may be positioned indoors, and the intake hood system **102** may receive indoor air containing dust, dirt, and/or smoke. As air passes through the intake hood system **102**, as indicated by arrow **104**, the air passes through air filters **106** and **108** positioned within the intake hood system **102**. The air filters **106** and **108** may be any filters suitable for filtering air, such as filters configured to remove dust, dirt, debris, smoke, and other particles from air. As explained in greater detail below, the air filters **106** and **108** are disposed at an inlet of the intake hood system **102**, such as across the inlet, to facilitate access to, insertion, and/or replacement of the air filters **106** and **108**.

(45) As illustrated, the intake hood system **102** is coupled to a side **110** of a housing **112** of the HVAC system **100**. After passing through the intake hood system **102**, air enters the housing **112**, as indicated by arrow **114**. The housing **112** may direct air toward a heating system, a cooling system, an additional filtering system, or any combination thereof, of the HVAC system **100**. For example, if the HVAC system **100** is operating in a heating operating mode, air may be directed from the intake hood system **102** toward the heating system. If the HVAC system **100** is operating in a cooling operating mode, air may be directed from the intake hood system **102** toward the cooling system. After passing through the heating system, the cooling system, the additional filters, and/or other portions of the HVAC system **100**, the air may be supplied to a building or other space conditioned by the HVAC system **100**.

(46) Air from the conditioned space may also be received by the HVAC system **100** as return air. For example, a flow of return air may enter the housing **112** of the HVAC system **100** and may be directed by a baffle **116** toward a vent **118** to exit the HVAC system **100** via the vent **118**, as indicated by arrow **120**. In some embodiments or modes of operation, the baffle **116** may direct the return air through the HVAC system **100** for conditioning before supplying the air to the conditioned space again. As illustrated, the vent **118** is coupled to the side **110** of the housing **112** and is positioned adjacent to the intake hood system **102**. In other embodiments, the intake hood system **102** and/or the vent **118** may be positioned elsewhere relative to the housing **112**, such as on different sides **110** of the housing **112**.

(47) FIG. **6** is a perspective view of an embodiment of the intake hood system **102**, illustrating an installed configuration of the intake hood system **102** and the air filters **106** and **108**. As illustrated, the intake hood system **102** includes an intake hood **140** configured to receive and direct air into the HVAC system **100**. For example, air may pass through the air filters **106** and **108** disposed within or at an inlet **142** of the intake hood **140**, as indicated by arrows **104**. The inlet **142** is at least partially defined by side panels **144** of the intake hood **140** that are disposed at ends **146** or lateral sides of the intake hood system **102**. For example, each side panel **144** may define a boundary

portion of the inlet **142**, such as a side boundary portion. In the illustrated embodiment, the air filter **106** includes a plurality of air filter segments **148** disposed between the side panels **144**, and the air filter **108** includes a plurality of air filter segments **150** disposed between the side panels **144**. Further, the intake hood **140** includes a top panel **152** disposed between and coupled to the side panels **144**. Each of the side panels **144** and the top panel **152** are configured to direct air received through the air filters **106** and **108** through the intake hood system **102** and toward an interior of the housing **112**.

(48) As illustrated, the intake hood system **102** includes a support **160** disposed between the side panels **144** and aligned with a boundary of the inlet **142**. In certain embodiments, the support **160** may be interference fit between the side panels **144** and/or disposed adjacent to the housing **112** of the HVAC system **100**. The support **160** is configured to engage with and/or capture an end **162** of the air filter **106**. The intake hood system **102** also includes an intermediate bracket **164** having supports **166** and **168** and disposed between the side panels **144**. The support **166** is configured to engage/capture an end **170** of the air filter **106** generally opposite of the end **162**, and the support **168** is configured to engage/capture an end **172** of the air filter **108**. As such, the intermediate bracket is disposed between the air filters **106** and **108**. Additionally, the intake hood system **102** includes a support **174** disposed between the side panels **144** and configured to engage/capture an end **176** of the air filter **108** generally opposite of the end **172**. As illustrated, each of the supports **160**, **166**, **168**, and **174** are C-channels configured to engage and/or capture respective ends of the air filters **106** and **108**. In certain embodiments, the supports **160**, **166**, **168**, and/or **174** may be other suitable supports configured engage ends of the air filters **106** and/or **108**, such as L-brackets, angular tubing, circular tubing, or other suitable channels, tubing, and/or brackets. In certain embodiments, the intermediate bracket **164** may be an I-bracket having and/or formed by the supports **166** and **168**. For example, each of the supports **166** and **168** may be C-channels that open toward opposite directions and that form an “I” shape.

(49) As described in greater detail below, the intake hood system **102** facilitates improved access to and/or installation/replacement of the air filters **106** and **108**. For example, during an installation process, the air filter **106** may slide into lateral supports, such as C-channels, of the side panels **144** and into the support **160**. The intermediate bracket **164** may then be inserted between the side panels **144**, such that the support **166** captures the end **170** of the air filter **106** and secures the air filter **106** within the intake hood system **102**. The air filter **108** may then be inserted into the support **168** and between the side panels **144**, and the support **174** may be inserted into the intake hood system **102**, such that the support **174** captures the end **176** of the air filter **108**. Thereafter, lock bars **178** of the intake hood system **102** may be coupled to the side panels **144** to secure the air filter **108** with the intake hood **140** of the intake hood system **102**. In this manner, the intake hood system **102** may be configured for quick assembly/disassembly and replacement/installation of the air filters **106** and **108**. As illustrated, the lock bars **178** are coupled to the side panels **144** of the intake hood **140** along a length of the intake hood **140** extending in a first direction that is transverse to a second direction in which the supports **160**, **166**, **168**, and **174** extend. In certain embodiments, the lock bars **178** may be coupled to the side panels **144** in other positions, such as extending generally in a common direction with the supports **160**, **166**, **168**, and **174**.

(50) FIG. 7 is a perspective view of an embodiment of intake hood system **102**, illustrating installation of the air filter **106** and the support **160** with the intake hood **140**. The side panels **144** of the intake hood **140** include lateral supports **200** that extend along a respective end **201** of each side panel **144** along the inlet **142** of the intake hood **140**. The lateral supports **200** are configured to capture lateral sides **202**, such as opposite lateral sides, of the air filter **106** when the air filter **106** is disposed within the inlet **142** of the intake hood **140**. From the illustrated position outside the intake hood **140**, the air filter **106** is configured to slide into the intake hood **140** via the lateral sides **202** and along the lateral supports **200** until the end **162** of the air filter **106** is received by and abuts the support **160**. As illustrated, a length of the support **160** extends transverse to

corresponding lengths of the lateral supports **200**. In other embodiments, the support **160** and/or the lateral supports **200** may extend in a common direction or in other suitable directions to engage the air filter **106**.

(51) As illustrated, the lateral supports **200** are formed by flanges **204** that are integral to the side panels **144** and flanges **206** that are fastened and/or secured to the side panels **144** in an interior region of the intake hood **140**. For example, each side **202** of the air filter **106** may be disposed between the flanges **204** and **206** of one of the side panels **144**. The flanges **204** and **206** form C-channels of the lateral supports **200** that capture/engage the lateral sides **202** of the air filter **106**, such that the lateral supports **200** may be lateral channels. In certain embodiments, the flanges **204** may be separate components that are fastened and/or secured to the side panels **144**, and/or the flanges **206** may be integral to the side panels **144**. In some embodiments, the flanges **204** and/or **206** may be inner flanges of the side panels **144**. As mentioned above, each of the flanges **204** and **206** is disposed along interiors **207** of the side panels **144**. As illustrated, the flanges **204** extend approximately one half of a length **210** of the side panels **144**, and the flanges **206** extend along a majority of the length **210** the side panels **144**. When inserting the air filter **106** into the intake hood **140**, the sides **202** of the air filter **106** may first abut the flanges **206**. Then, the air filter **106** may slide into the lateral supports **200** and abut both the flanges **204** and the flanges **206**. As such, the lateral supports **200** enable quick and efficient insertion and removal of the air filter **106** into and from the intake hood **140**.

(52) FIG. **8** is a perspective view of an embodiment of the intake hood system **102**, illustrating installation of the intermediate bracket **164** with the intake hood system **102**. As described herein, the support **166** of the intermediate bracket **164** is configured to capture the end **170** of the air filter **106**, and the support **168** of the intermediate bracket **164** is configured to capture the end **172** of the air filter **108**. From the illustrated position of the intermediate bracket **164** outside of the intake hood **140**, the intermediate bracket **164** is configured to slide onto the end **170** of the air filter **106** and between the side panels **144**. In certain embodiments, the intermediate bracket **164** may be at partially disposed within the lateral supports **200**, similar to the air filter **106**. For example, the intermediate bracket **164** may at least partially abut the flanges **204** and/or the flanges **206** when coupled to the air filter **106**. In some embodiments, the support **166** of the intermediate bracket **164** may be disposed within the lateral supports **200**, but the support **168** of the intermediate bracket **164** may not be disposed within the lateral supports **200**. As such, the intermediate bracket **164** and/or the support **166** may be interference fit between the side panels **144** and/or between the flanges **204** and **206** of each lateral support **200**.

(53) In certain embodiments, the intermediate bracket **164** may be coupled to the air filter **106** prior to insertion of the air filter **106** into the intake hood **140**. For example, the end **170** of the air filter **106** may be inserted into the support **166** of the intermediate bracket **164**, and both the air filter **106** and the intermediate bracket **164** may be inserted into the intake hood **140**, such as by sliding the air filter **106** and the intermediate bracket **164** into the lateral supports **200** until the end **162** of the air filter **106** is received by and abuts the support **160**.

(54) FIG. **9** is a perspective view of an embodiment of the intake hood system **102**, illustrating installation of the air filter **108** with the intake hood system **102**. From the illustrated position of the air filter **108** outside of the intake hood **140**, the end **172** of the air filter **108** is configured to slide into the support **168** and be disposed between the side panels **144**. For example, sides **220** of the air filter **108** may abut the flanges **206** of the side panels **144** while disposed between the side panels **144**.

(55) In certain embodiments, the air filter **108** may be coupled to the intermediate bracket **164** prior to insertion into the intake hood **140**. For example, the end **172** of the air filter **108** may be inserted into the support **168** of the intermediate bracket **164** outside of the intake hood **140**, and both the air filter **108** and the intermediate bracket **164** may be inserted into the intake hood **140**, such as by sliding the support **166** of the intermediate bracket **164** onto the end **170** of the air filter **106** and in

between the side panels **144**. In some embodiments, the air filter **106**, the intermediate bracket **164**, and the air filter **108** may be assembled prior to insertion into the intake hood **140** and thereafter inserted as a filter and bracket assembly into the intake hood **140**.

(56) FIG. **10** is a perspective view of an embodiment of the intake hood system **104**, illustrating installation of the support **174** with the intake hood system **102**. As illustrated, the support **174** is disposed outside the intake hood **140**. The support **174** is configured to slide onto and capture the end **176** of the air filter **108**. In certain embodiments, the support **174** may be coupled to the end **176** of the air filter **108** prior to insertion of the air filter **108**, the intermediate bracket **164**, and/or the air filter **106** into the intake hood **140**. As such, the support **174** may be offset from the lateral supports **200** and the flanges **204** of the side panels **144**. For example, the end **176** of the air filter **108** may be inserted into the support **174**, and then the support **174**, the air filter **108**, the intermediate bracket **164**, and/or the air filter **106** may be simultaneously inserted into the intake hood **140**.

(57) FIG. **11** is a perspective view of an embodiment of the intake hood system **102**, illustrating installation of the lock bars **178** with the intake hood system **102**. From the illustrated position of the lock bars **178** apart from the intake hood **140**, the lock bars **178** may be coupled to flanges **230** of the side panels **144** to secure the air filter **108** within the inlet **142** of the intake hood **140**. The flanges **230** extend outwardly from the side panels **144** and the inlet **142**. As described in greater detail below, each lock bar **178** may be coupled to one of the flanges **230** via fasteners. Once secured to the intake hood **140**, the lock bars **178** are configured to abut the supports **166**, **168**, and/or **174** to secure the air filter **108** within the inlet **142** of the intake hood **140**. For example, the air filter **108** may be disposed/captured between the flanges **206** of the side panels **144** and the lock bars **178**. As such, in certain embodiments, the flanges **206** and the lock bars **178** may cooperatively form lateral supports configured to capture the sides **220** of the air filter **108**. In certain embodiments, the supports **166**, **168**, and/or **174** and/or the air filter **108** may be disposed laterally between the flanges **230** of the side panels **144**. Referring back to FIG. **6**, the lock bars **178** are coupled to the flanges **230** of the side panels **144** such that lock bars **178** secure the air filter **108** within the inlet **142** of the intake hood **140**. As illustrated, the lock bars **178** are channeled members, such as C-channels. In other embodiments, the lock bars **178** may be any shape suitable for abutting the supports **166**, **168**, and/or **174** to secure the air filter **108** within the inlet **142** of the intake hood **140** when coupled to the flanges **230**, such as flattened plates, angular brackets, tubes, and other suitable shapes.

(58) FIG. **12** is an expanded perspective view of an embodiment of the intake hood system **102**, illustrating the lock bar **178** coupled to the intake hood **140**. As shown, the lock bar **178** is coupled to the flange **230** of the side panel **144** via a fastener assembly **240**. The fastener assembly **240** is configured to extend through the lock bar **178** and the flange **230** to secure the lock bar **178** to the intake hood **140**. Additionally, the fastener assembly **240** may be reusable such that at least a portion of the fastener assembly **240** may be removed and reinstalled multiple times to enable repeated removal and reinstallation of the lock bar **178** and access to the air filters **106** and **108** without replacement of the fastener assembly **240**. As illustrated, the fastener assembly **240** includes a fastener **242**, such as a screw or a bolt, and a nut **244**, such as a thumb nut or a rivet nut, threadingly engaged with one another. The nut **244** may be removed from and reattached to the fastener **242** multiple times without damaging the fastener assembly **240**, the lock bar **178**, or the flange **230**. For example, the nut **244** may be unscrewed from the fastener **242** and/or attached to the fastener **242** by hand or tool without damaging threads of the fastener **242** or the nut **244**, and the fastener assembly **240** may be removed from and reinserted through the lock bar **178** and the flange **230** without damaging the lock bar **178** or the flange **230**. In some embodiments, one or more portions of the fastener assembly **240** may be fixed to the lock bar **178** and/or the flange **230**. For example, the fastener **242** may be rigidly fixed to the flange **230** and configured to extend from the flange **230**. In such embodiments, the lock bar **178** may be inserted over the fastener **242** and

the nut **244** attached to the fastener **242** to secure the lock bar **178** to the flange **230**. In certain embodiments, the nut **244** may be rigidly fixed to the lock bar **178**, such as integrally formed with and/or press fit to the lock bar **178**, such that the fastener **242** may be inserted through the flange **230** and the lock bar **178** and into the nut **244** to secure the lock bar **178** to the flange **230**.

(59) As illustrated, the lock bar **178** includes a portion **246** that extends along and abuts the flange **230**. In certain embodiments, the portion **246** may extend along the entire flange **230**. The lock bar **178** also includes a portion **248** that extends along the air filter **108**, the support **174**, the intermediate bracket **164**, and/or the flange **204** of the side panel **144**. The portion **248** enables the lock bar **178** to secure the air filter **108**, the support **174**, and/or the intermediate bracket **164** within the intake hood **140**. For example, while the lock bar **178** is secured to the intake hood **140**, such as via the portion **246** secured to the flange **230**, the lock bar **178** may also support and retain the air filter **108**, the support **174**, and/or the intermediate bracket **164** within the inlet **142** of the intake hood **140**, such as via the portion **248** that abuts the air filter **108**, the support **174**, the intermediate bracket **164**, and/or the flange **204**.

(60) FIG. **13** is an expanded perspective view of an embodiment of the intake hood system **102**, illustrating the lock bar **178** coupled to the flange **230** of the intake hood **140**. The lock bar **178** is secured to the flange **230** via the fastener assembly **240**, which extends through the portion **246** of the lock bar **178**. As described above, the portion **248** of the lock bar **178** may extend along and generally abut the air filter **108**, the support **174**, the intermediate bracket **164**, and/or the flange **204** of the side panel **144**. As illustrated, the lock bar **178** extends along and abuts the air filter **108**, the support **168** of the intermediate bracket **164**, and/or the flange **204**. In certain embodiments, the portion **248** may extend along and abut the air filter **108** without directly abutting other components. In some embodiments, the portion **248** may extend along and abut both of the supports **166** and **168** of the intermediate bracket **164**. In further embodiments, the portion **248** of the lock bar **178** may extend along and abut the air filter **106** and/or a majority of the flange **204** of the side panel **144**. As such, the lock bar **178**, via the portions **246** and **248**, is configured to secure the air filters **106** and/or **108**, the support **174**, and/or the supports **166** and/or **168** of the intermediate bracket **164** within the intake hood **140**. In the illustrated embodiments, each lock bar **178** is coupled to one of the flanges **230** via two fastener assemblies **240**. In some embodiments, one or both lock bars **178** may be coupled to the flanges **230** via more or fewer fastener assemblies **240**, such as by one fastener assembly **240**, three fastener assemblies **240**, four fastener assemblies **240**, or six fastener assemblies **240**.

(61) FIG. **14** is a flow diagram of an embodiment of a process **260** for assembling the intake hood system **102**. At block **262**, the air filter **106** is inserted into the lateral supports **200**, such as lateral channels, and into the support **160**, such as a C-channel. For example, the sides **202** of the air filter **106** may be inserted between the flanges **204** and **206** that form the lateral supports **200** and slid into the lateral supports **200** until the end **162** of the air filter **106** abuts the support **160**. In certain embodiments, one or more of the plurality of air filter segments **148** of the air filter **106** may be individually inserted into the intake hood **140**, such as into the lateral supports **200** and/or into the support **160**.

(62) At block **264**, the intermediate bracket **164** is installed, such that the support **166** captures and fits over the end **170** of the air filter **106**. As described above, the support **166** of the intermediate bracket **164** may be coupled to the air filter **106** prior to insertion of the air filter **106** into the intake hood **140**. For example, the air filter **106** may first be inserted into the support **166**, and both the air filter **106** and the intermediate bracket **164** may be inserted/installed to the intake hood **140**.

(63) At block **266**, the air filter **108** is inserted into the support **168** of the intermediate bracket **164**, which is configured to capture the end **172** of the air filter **108**. In certain embodiments, the air filter **108** may be coupled to the intermediate bracket **164** and/or the air filter **106** prior to insertion of the intermediate bracket **164** and/or the air filter **106** into the intake hood **140**. In some embodiments, one or more of the plurality of air filter segments **150** of the air filter **108** may be

individually inserted into the intake hood **140**, such as into the support **168** of the intermediate bracket **164**.

(64) At block **268**, the support **174** is inserted over the air filter **108**. For example, the support **174** is configured to capture the end **176** of the air filter **108**. In certain embodiments, the support **174** may be coupled to the air filter **108** prior to insertion of the air filter **108** into the intake hood **140**.

(65) At block **270**, the lock bars **178**, such as lock plates, are installed/coupled to the intake hood **140** to secure the air filter **108**, the support **174**, and/or the intermediate bracket **164** within the inlet **142** of the intake hood **140**. For example, the lock bars **178** may be positioned adjacent to the side panels **144**, such that the lock bars **178** abut the flanges **230** of the side panels **144**, and at least a portion each fastener assembly **240** may be inserted into/through the lock bars **178** to secure the lock bars **178** to the intake hood **140**. As such, after inserting the air filters **106** and **108**, the intermediate bracket **164**, and the support **174** into the inlet **142** of the intake hood **140**, the lock bars **178** may be fastened to the side panels **144** to secure the air filters **106** and **108**, the intermediate bracket **164**, and the support **174** within the inlet **142**.

(66) FIG. **15** is a flow diagram of an embodiment of a process **280** for disassembling the intake hood system **102**. At block **282**, the lock bars **178** are removed from the intake hood **140** to enable removal of the air filter **108**, the support **174**, and/or the intermediate bracket **164** from the intake hood **140**. For example, at least a portion of the each fastener assembly **240** may be removed to enable removal/uninstallation of the lock bars **178**. As described above, the nut **244** of each fastener assembly **240**, such as a thumb nut, may be unscrewed from each respective fastener **242** to enable removal of the lock bars **178**.

(67) At block **284**, the support **174** is removed from the end **176** of the air filter **108**. At block **286**, the air filter **108** is removed from the support **168** of the intermediate bracket **164**. In certain embodiments, the support **174** and the air filter **108** may be removed at substantially the same time, such as simultaneously, from the intake hood **140**. In some embodiments, one or more of the plurality of air filter segments **150** of the air filter **108** may be individually removed from the intake hood **140**, such as from the support **168** of the intermediate bracket **164**.

(68) At block **288**, the intermediate bracket **164** having the supports **166** and **168** is removed from the intake hood **140**. For example, after removing the air filter **108**, the intermediate bracket **164** may be pulled off the air filter **106**. In certain embodiments, the intermediate bracket **164** may be removed from the intake hood **140** at the same time as the air filter **108**, such that the air filter **108** is still disposed within the support **168** of the intermediate bracket **164** when the air filter **108** and the intermediate bracket **164** are removed from the intake hood.

(69) At block **290**, the air filter **106** is removed from the intake hood **140**. For example, the end **162** of the air filter **106** may be removed from the support **160**, and the sides **202** of the air filter **106** may be slid along and out of the lateral supports **200** until the air filter **106** is removed from the intake hood **140**. In certain embodiments, the air filter **106** may be removed from the intake hood **140** at the same time as the intermediate bracket **164** such that the end **170** of the air filter **106** is still disposed within the support **166** of the intermediate bracket **164** when the intermediate bracket **164** and the air filter **106** are removed from the intake hood **140**. In some embodiments, one or more of the plurality of air filter segments **148** of the air filter **106** may be individually removed from the intake hood **140**, such as from the support **160**.

(70) Accordingly, the present disclosure provides systems and methods for an intake hood system generally positioned at an inlet of an HVAC system. The intake hood system includes air filters configured to remove dirt, debris, and other particles from air directed through the intake hood system and into, for example, a housing of an HVAC unit. The disclosed techniques enable quick and efficient access to the air filters, such as access by service personnel, to quickly service and/or replace the air filters. For example, the intake hood system may include supports, brackets, and/or side panels that capture the air filters. The supports, the brackets, and/or the side panels may be easily disassembled/uninstalled to remove the air filters and assembled/installed to insert/replace

the air filters. As such, the systems and methods described herein improve serviceability of the HVAC system and facilitate installation of air filters.

(71) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

(72) The specific embodiments described above have been shown by way of example, and it should be understood that these embodiments may be susceptible to various modifications and alternative forms. It should be further understood that the claims are not intended to be limited to the particular forms disclosed, but rather to cover all modifications, equivalents, and alternatives falling within the spirit and scope of this disclosure.

Claims

1. An intake hood system for a heating, ventilation, and/or air conditioning (HVAC) unit, comprising: an intake hood having a first side panel and a second side panel forming an inlet of the intake hood, wherein the first side panel includes a first inner flange and a first outer flange, wherein the first inner flange at least partially defines a first lateral channel configured to capture a first side of an air filter, and the second side panel includes a second inner flange and a second outer flange, wherein the second inner flange at least partially defines a second lateral channel configured to capture a second side of the air filter, wherein the first inner flange and the second inner flange extend inwardly relative to the inlet and the first outer flange and the second outer flange extend outwardly relative to the inlet; a first support disposed between the first side panel and the second side panel such that a length of the first support is transverse to corresponding lengths of the first and second lateral channels, wherein the first support is configured to capture a first end of the air filter and align with the inlet; and a second support disposed between the first side panel and the second side panel and configured to capture a second end of the air filter.
2. The intake hood system of claim 1, comprising an intermediate bracket having the second support and a third support configured to capture a first end of an additional air filter at the inlet.
3. The intake hood system of claim 2, wherein the first support, the second support, the third support, and a fourth support configured to capture a second end of the additional air filter each respectively form a C-channel.
4. The intake hood system of claim 3, wherein the fourth support is configured to be disposed between the first outer flange and the second outer flange.
5. The intake hood system of claim 4, comprising a lock plate configured to couple to the first outer flange or the second outer flange and abut the third support and the fourth support to secure the additional air filter with the intake hood.
6. The intake hood system of claim 4, wherein the first inner flange of the first lateral channel and the first outer flange extend along an end of the first side panel, and the second inner flange of the second lateral channel and the second outer flange extend along an end of the second side panel.
7. The intake hood system of claim 4, wherein the first side panel includes a third inner flange extending along an interior surface of the first side panel, the second side panel includes a fourth inner flange extending along an interior surface of the second side panel, the first inner flange and the third inner flange form the first lateral channel, and the second inner flange and the fourth inner flange form the second lateral channel.
8. The intake hood system of claim 1, wherein the first support is configured to be interference fit between the first side panel and the second side panel.

9. The intake hood system of claim 1, wherein the first side panel and the second side panel are configured to couple the intake hood to a housing of the HVAC unit.

10. The intake hood system of claim 1, wherein the first inner flange and the first outer flange extend in a common direction along a length of the first side panel, and the second inner flange and the second outer flange extend in the common direction along a length of the second side panel.

11. An intake hood system for a heating, ventilation, and/or air conditioning (HVAC) unit, comprising: a first side panel and a second side panel forming an intake hood, the first side panel and the second side panel each having a first inner flange and a second inner flange; a first C-channel configured to interference fit between the first inner flange and the second inner flange of each of the first side panel and the second side panel and configured to capture a first end of an air filter; an intermediate bracket configured to interference fit between the first inner flange and the second inner flange of each of the first side panel and the second side panel and having a second C-channel and a third C-channel, wherein the second C-channel is configured to capture a second end of the air filter, and the third C-channel is configured to capture a first end of an additional air filter; and a fourth C-channel configured to capture a second end of the additional air filter, wherein the fourth C-channel is configured to be offset from the first inner flange and the second inner flange of each of the first side panel and the second side panel.

12. The intake hood system of claim 11, comprising a first lock plate configured to couple to the first side panel and a second lock plate configured to couple to the second side panel, wherein each of the first lock plate and the second lock plate is configured to abut the fourth C-channel and the intermediate bracket to secure the additional air filter at an inlet of the intake hood.

13. The intake hood system of claim 11, wherein the first inner flange and the second inner flange of the first side panel form a first lateral C-channel configured to capture the air filter, and the first inner flange and the second inner flange of the second side panel form a second lateral C-channel configured to capture the air filter.

14. The intake hood system of claim 11, comprising the air filter and the additional air filter, wherein the air filter and the additional air filter are configured to be disposed at and across an inlet of the intake hood.
